Satellite Communication System (S.O.O. 6298) – General Data

1. General

Provisions are provided for the Satellite Communication System which consists of aircraft flight tracking and two-way telephone communication.

2. Description

The Satellite Communications System provides the following functions:

1 Flight-tracking which provides automatic position reporting at configurable intervals between the aircraft and ground-based personal computer, via the Iridium Low Earth Orbit (LEO) Satellite Communications Transceiver.

2 Satellite telephone audio conversion via the Iridium Satellite Communications System. Communications is through the COM 5 transmit channel on the pilot or co-pilot's audio panels. Control, including dialing is via the RC6000/A panel in the center pedestal.

3 Short text messaging via the RC6000/A panel in the center pedestal.

The system includes the following components:

1 RC6000/A control panel in the center pedestal.

2 S200 SkyNode in the forward avionics bay.

3 Low profile GPS/Iridium antenna on the upper fuselage at station 246.5.

4 A 3 Amp circuit breaker in the avionics CB panel labeled either "SATELLITE PHONE" or "SAT COM", typically located in position 4N.

The Satellite Communications System has no power switch. The system operates whenever power is present on the RH bus, and will start transmitting position reports when the GPS detects roundspeed.

As delivered, the SkyNode is set to default reporting settings. However, it is possible to modify the system configuration and settings. Some settings, such as the GPS Position Recording Interval (length of time between the SkyNode's processing and storing of a GPS position) and Connection Interval (length of time between regularly scheduled transmissions of position data), can be adjusted by the user from a Latitude Web Sentinel™ account.

Other settings such as OOOI (Out, Off, On, In) reporting, and Latitude Messaging API can be altered remotely by Latitude Technologies Corporation.

Contact Latitude Technologies Corporation for system activations and account information.

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Device Activations: 1-250-475-0203; activations@latitudetech.com

Table 1 Control Buttons

Home	The “HOME” key brings the user back to RC6000/A’s main menu screen
Menu	No function
Soft Keys	The RC6000/A has three keys that are used as contextual buttons: “S1”, “S2” and “S3”. The functions and labels of these keys change depending on the operating mode of the control panel.
Function Keys	No function
Alphanumeric Keys	There are 10 alphanumeric keys similar to those found on a telephone. These are used for dialing telephone numbers or entering text.
Arrow and Return Keys	The arrow keys and the return key (aka the enter key; “ ↵ ”) are used to navigate the RC6000/A menus. Use the arrow keys to navigate with the menus and use the return key to select a menu item.
Mark and DN/CLR	The “Mark” (“Mark Target”, also labeled “#”) and “DN/CLR” (“Down & Clear”, also labeled “*”) keys can be used as signaling inputs; pressing one of these keys triggers the SkyNode SAT COM to immediately generate and transmit a GPS position report with a special reason code that will be displayed on the Web Sentinel ground based computer. (Note 1)
May Day	The “MAY DAY” key can be used to place the SkyNode in Emergency Tracking mode. This means that the SkyNode will override the configured reporting interval and begin transmitting specially flagged GPS position reports every 10 seconds. (Notes 1, 2 and 3)

Note

1 Special feature, contact Latitude Technologies.

2 The destination to where the emergency messages are sent is determined by the SkyNode SAT COM equipment that the RC6000/A is connected with. At minimum, the SAT COM must have an active satellite network account and have been configured for emergency reporting.

3 Emergency Tracking MAY DAY mode is not part of the emergency locator transmitter. It is a supplementary capability that sends position reports to the company Web Sentinel base station at faster time intervals than for normal position reports.

3. Equipment and Materials

- Cleaner MIL–C–2576
- Isopropanol
- Digital Multimeter
- Milliohm meter (capable of measuring <3.0 milliohms)
- Personal Computer (PC) with Web Sentinel Software, internet connection and a web browser

4. Cleaning

The antenna may be cleaned only with the approved cleaning products listed in Para 3. Refer to 12-30-01, Aircraft External Cleaning – General Data for cleaning procedures. Limit cleaning to wiping with moistened cloths.

5. Adjustment/Test

A. Function Test of the SAT COM System

This test requires:

- A functioning ground based PC as described in Para 3., and radio or mobile phone communications between the cockpit and the ground based PC.

- A registered account with Latitude Technologies.
- That the forward upper panel of the forward avionics bay be temporarily removed to permit observation of the LED lights on the SkyNode.

1) General

   a) Position the aircraft to receive GPS and Iridium signals.

   b) Switch aircraft power ON.

   c) Open the SAT COM (N4) circuit breaker on the avionics circuit breaker panel and ensure that the control panel display, in the center pedestal is OFF and that the LED's on the SkyNode transceiver in the nose are OFF.

   d) Close the SAT COM (N4) circuit breaker on the avionics circuit breaker panel and ensure that the control panel display in the center pedestal illuminates and that the LED's on the SkyNode transceiver in the nose are ON (refer to troubleshooting procedures in Para 7., for a description of the LED blink patterns).

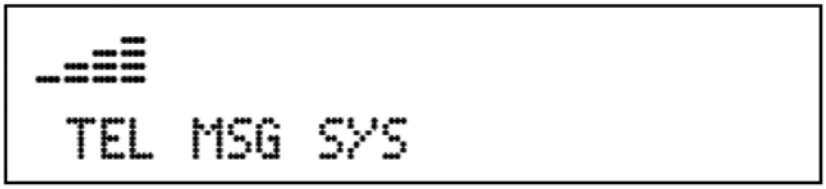
2) Power Up and Position Report Check

   a) During startup, the RC6000/A control panel will display an animated cursor moving right and left on the display's bottom line, while the top line will alternate between displaying the following two lines:

```
LATITUDE RC6000/A FW#.##
** SEARCHING FOR SkyNode **
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   Where "#.##" is the firmware version currently installed on the RC6000/A.

   b) After the SkyNode is fully initialized and communication is established with the RC6000/A, the home screen will appear with TEL, MSG and SYS above select buttons S1, S2 and S3 respectively:



   The home screen is the starting point for access all other areas of the menu. From any other screen, or during any function operation, it is possible to return to the home screen by pressing the HOME button.

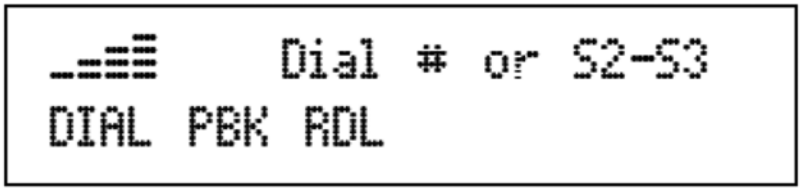
   c) There are two LED indicators on the left. The bottom LED has no function. The top LED indicates the status of short text messages received by the system. OFF if there are no messages in the inbox, BLINKING if there are messages in the inbox that are unused, and ON if there are messages that have all been read.

   d) Adjust screen brightness by pressing button "S3" (labeled "SYS"); to display the System Menu. Then press "S3" again, (labeled "BRT") from the System Menu. Adjust the screen brightness by using soft keys "S2" and "S3" or the arrow keys.

   e) After the SkyNode is fully initialized, confirm that the aircraft position is being displayed on the Web Sentinel display on the ground based PC. Confirm that the displayed position is reasonable.

3) Outbound Satellite Telephone Call Check

a) To use the satellite telephone, start in the Home Screen then press “S1” (labeled “TEL”); this will display the Main Telephone Screen.



Note When dialing a phone number, remember that all calls placed from an Iridium telephone are considered international calls and must begin with an international dialing code (for example, “001” for North America).

b) Wear a headset, and select the SAT COM transceiver (usually COM 5) on the cockpit audio panel.

c) From the Main Telephone Screen, use the number keys to dial a phone number. The number you are entering will be displayed on the top line of the display. Once you have finished dialing, press “S1” (labeled “DIAL”) to initiate the call.

Note Sending Dual-Tone Multi-Frequency (DTMF) tones; there are often instances when you must press keys during a voice call, such as remotely checking voicemail and navigating telephone menu systems. When you press a key during the call, the bottom right corner of the screen will display the text:

Sending ‘#’ tone

Where “#” is the key that was pressed (i.e. 1–9, #, *). You must wait until this text disappears before pressing another key.

d) When the call is answered, use the transmit button when talking, and release the transmit button when receiving.

e) To end a phone call that is in progress (i.e. “hang up”), press “S1” (labeled “END”).

4) Inbound Satellite Telephone Call Check

a) When a satellite call is received, the top line of the screen will display the flashing text:

***** Incoming Call *****

b) An aural “ringing” alert will be heard if the SAT COM transceiver (usually COM 5) is selected on the aircraft audio panel when the inbound call is received.

c) To answer the call select the SAT COM transceiver (usually COM 5) on the cockpit audio panel and press “S2” (labeled “ANS”) on the SAT COM control panel.

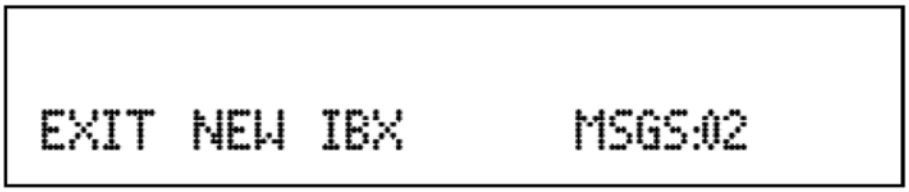
d) When the call is answered, use the transmit button when talking, and release the transmit button when receiving.

e) To end a phone call that is in progress (i.e. “hang up”), press “S1” (labeled “END”).

5) Messaging Send Check

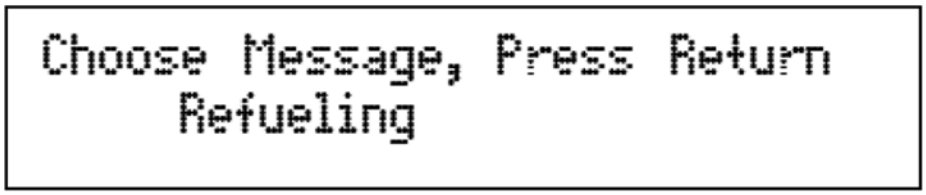
Sent messages are viewed on the ground in Web Sentinel. Forwarding to user-configured emails can also be setup in your Web Sentinel administrator account.

a) Start in the Home Screen, then press “S2” (labeled “MSG”); to display the Main Messaging Screen.



b) Press “S2” (labeled “NEW”), the screen will display the first of 9 canned messages on the bottom line. There are two types of canned message; Static and Hybrid. Canned messages are configured remotely from a Web Sentinel administrator account.

c) Use the arrow keys to navigate to your desired message, and then press return to select the message.



d) When a Static Canned Message is selected, you will be prompted to confirm that you want to send the message (soft keys “S1” and “S3” will be flashing “NO” and “YES”); simply press “YES” and the message will transmit (pressing “NO” will return you to the Main Messaging Screen).

e) Confirm that the Web Sentinel PC receives the message.

6) Messaging Receive Check

a) Send a message from the Web Sentinel PC to the aircraft.

b) When a new text message is received, the LED (green) at the left-side of the display’s top line will begin to blink, and the word “*MESSAGE*” will begin to blink on the bottom line of the display. These indications will continue until all messages in the inbox have been read.

c) Navigate to the inbox from the Main Messaging Screen by pressing “S3” (labeled “IBX”).

d) Using the arrow keys, navigate to the desired message, then press “S2” (labeled “REP”) to be taken to the New Message Screen or “S3” (labeled “DEL”) to delete the message (you will be prompted to confirm deletion).



e) A message is marked as read after being displayed for a few seconds or after one complete scroll across the screen in the case of longer messages.

6. Removal/Installation

A. Combined GPS/Iridium/Wi-Fi Antenna

1) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Open the SAT COM (N4) circuit breaker on the avionics circuit breaker panel and fit safety clip. If S.O.O. 6288 is installed, open the ECTM/FOQA (Q4) circuit breaker and fit safety clip.

c) Remove the four (4) screws securing the antenna to the airframe.

d) Carefully cut the sealant from around the base of the antenna.

e) Carefully lift the antenna clear of the fuselage until there is access to the coaxial connections.

f) Disconnect the coaxial connectors.

Note Stow connectors as required.

g) Remove the antenna.

2) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure the SAT COM (N4) circuit breaker is opened and fitted with safety clips. When S.O.O. 6288 is installed, ensure ECTM/FOQA (Q4) circuit breaker is opened and fitted with safety clip.

c) Remove non-conductive material (e.g. anodic films, primer, paint, etc) from the electrical bond contact area, in accordance with 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General Data, then apply chemical conversion coating (C10), in accordance with 20-30-02, Chemical Conversion Coatings (Alodine) – Manual Application, within 8 hours of stripping high resistance films.

d) Solvent clean the electrical contact bond areas immediately before installation. (Refer to 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General Data).

e) Move the antenna close to the fuselage and connect the coaxial connectors.

f) Secure the antenna to the fuselage using the four (4) mounting screws.

g) Solvent clean the exposed areas of the airframe around the antenna. (Refer to 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General Data).

h) Touch up exposed areas of the airframe around the antenna, if required, with F19 primers by brush within 24 hours of the antenna installation. Ensure that the primer re-finishing provides complete coverage of the affected area.

i) Do a resistance measurement between the antenna and the airframe. Resistance must be less than 3 milliohms, in accordance with 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General Data.

j) If the affected area requires paint touch up, brush coat the area with the top coat paint system matching the existing.

k) Fillet seal around the base of the antenna with MIL–PRF–81733 sealant, in accordance with 20-50-01, Weather Sealing – General Data. The aircraft will be ready to fly 24 hours after application of the sealant.

l) Remove the safety clip and close the SAT COM (N4) circuit breaker. When S.O.O. 6288 is installed, remove the safety clip and close the ECTM/FOQA (Q4) circuit breaker.

m) Do a function test of the SAT COM system per Para 5.A.1)

B. Dialer

1) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Open the SAT COM (N4) circuit breaker on the avionics circuit breaker panel and fit safety clip. If S.O.O. 6288 is installed, open the ECTM/FOQA (Q4) circuit breaker and fit safety clip.

c) Unscrew the built-in DZUS fasteners in the dialer unit.

d) Carefully pull the dialer from the center column until there is access to the wire connector.

e) Disconnect the wire connector.

Note Stow connectors as required.

f) Remove the dialer.

2) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure the SAT COM (N4) circuit breaker is opened and fitted with safety clip. When S.O.O. 6288 is installed, ensure the ECTM/FOQA (Q4) circuit breaker is opened and fitted with safety clip.

c) Move the dialer close to the centre column and connect the wire connector.

d) Carefully insert the dialer into the centre column and tighten the built-in DZUS fasteners.

e) Remove the safety clip and close the SAT COM (N4) circuit breaker. When S.O.O. 6288 is installed, remove safety clip and close the ECTM/FOQA (Q4) circuit breaker.

f) Do a function test of the SAT COM system as per Para 5.A.1).

C. SkyNode Unit

1) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Open the SAT COM (N4) circuit breaker on the avionics circuit breaker panel and fit safety clip. If S.O.O. 6288 is installed, open the ECTM/FOQA (Q4) circuit breaker and fit safety clip.

c) Disconnect all the connectors.

Note Stow connectors as required.

d) Remove the four (4) screws and washers holding the SkyNode in position on the top face of the upper, left avionics shelf in the FWD cargo compartment.

e) Remove the SkyNode.

2) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure the SAT COM (N4) circuit breaker is opened and fitted with safety clip. When S.O.O. 6288 is installed, ensure the ECTM/FOQA (Q4) circuit breaker is opened and fitted with safety clip.

c) Carefully position the SkyNode over the four (4) potted mounting holes on the top face of the upper, left avionics shelf in the FWD cargo compartment.

d) Install the four (4) screws and washers to secure the SkyNode in position.

e) Connect all the connectors.

f) Remove the safety clip and close the SAT COM (N4) circuit breaker. When S.O.O. 6288 is installed, remove safety clip and close the ECTM/FOQA (Q4) circuit breaker.

g) Do a function test of the SAT COM system as per Para 5.A.1).

7. Troubleshooting

A. Basic Checks

1) Ensure that the electrical system is ON (circuit breaker closed, LED on S200 SkyNode is active and control panel is illuminated).

2) Verify the SkyNode is secured and that all cables including antenna cables are in good condition and properly connected.

3) Verify that the antenna connectors at the antenna are secure and in good condition.

4) Verify that the antenna installation is clean and free of contaminants and corrosion.

5) Verify that the control panel is properly connected and properly secured into the instrument panel.

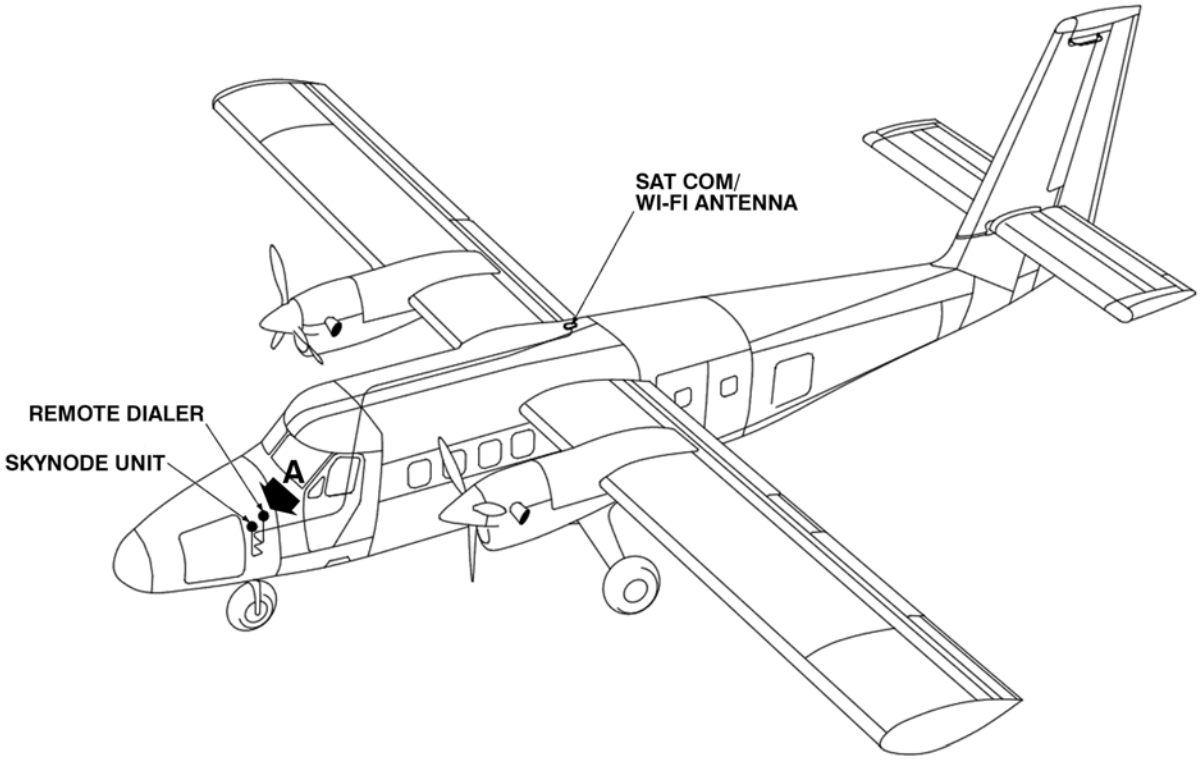
6) Refer to the Wiring Diagrams Manual (PSM 1-64-2W) for circuit breaker troubleshooting.

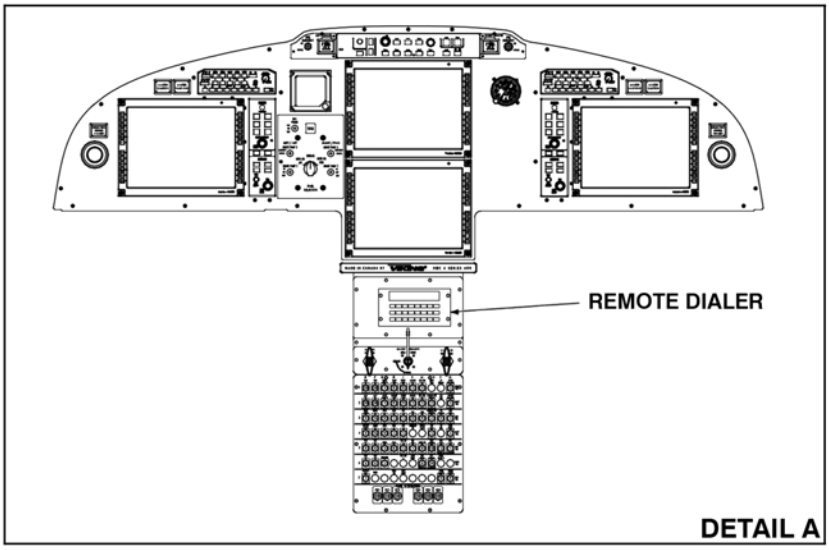
7) Investigate the LED blink pattern on the S200 SkyNode. Table 2 — Blink Pattern Meanings describes the meaning of the blink pattern.

Table 2 Blink Pattern Meanings

LED OBSERVATION	SYSTEM STATUS
LED On Solid	System running normally
Long On, Short Off	System is initializing (i.e. start up)
1 Blink Every 3 Seconds	No GPS satellite fix
2 Blinks Every 3 Seconds	No Iridium satellite fix
1 Blink in 3 Seconds, Then 2 Blinks in 3 Seconds	No Iridium network and No GPS fix
On for 1 Second Every 2 Seconds	Pass-Through Mode

8) Verify that an account with Latitude Technologies is active (refer to Contact Information in Para 2.).





Satellite Communication System Locations (S.O.O. 6298)
Figure 1

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